

Panel 3 - Time-marching in dynamics



Newmark integration

Given: $\mathbf{u}_{n-1}$, $\dot{\mathbf{u}}_{n-1}$, $\ddot{\mathbf{u}}_{n-1}$

Predict: $\mathbf{u}_n$ (by Point-DeepONet)

Update: $\dot{\mathbf{u}}_n$, $\ddot{\mathbf{u}}_n$

FE solver in Ω_{FE}

FEM solve with
interface BCs

Point-DeepONet

in Ω_{NO}

Predict $\mathbf{u}_n$ on
point cloud